INCH-POUND
MIL-PRF-62/2E
18 August 2020
SUPERSEDING
MIL-C-62/2D
18 December 2008

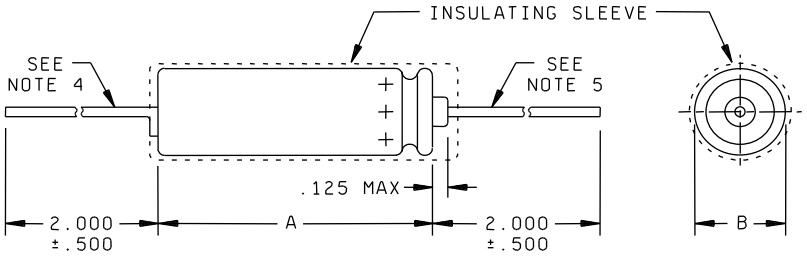
PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, ELECTROLYTIC (DC, ALUMINUM, DRY ELECTROLYTE, POLARIZED), STYLES CE12 AND CE13

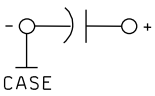
INACTIVE FOR NEW DESIGN
AFTER 1 JULY 1998

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-62.



STYLES
CE12 (uninsulated)
CE13 (insulated)



Inches	mm
.125	3.18
.500	12.70
2.000	50.80

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Capacitors with a length of 1.625 inches (41.28 mm) or greater or a diameter of 0.750 inch (19.05 mm) or greater are not intended to be mounted by their leads. They shall be provided with a supplementary means of mounting such as a tangential bracket or wraparound bands. (Family 1 (see MIL-PRF-62)).
4. The negative lead need not be offset.
5. Leads shall be .032 inch (0.81 mm) ± .002 inch (0.05 mm) (20 AWG) for units with a diameter of .625 inch (15.88 mm) or less and .040 inch (1.02 mm) ± .002 inch (0.05 mm) (18 AWG) for units with a diameter greater than .625 inch (15.88 mm).

FIGURE 1. Dimensions and configuration.

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TABLE I. Styles CE12 and CE13 capacitors.

Part or identifying number (PIN) 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 +10, -5 Hz, -40°C +0°C, -3°C	Dissipation factor	Case dimensions 2/ 3/	
						A	B
						± .062 (1.57)	± .031 (0.79)
	µf	Volts	µa	Ohms	Percent	Inches (mm)	Inches (mm)
CE1-C181C	180	5	45	194	40	1.125 (28.58)	.500 (12.70)
CE1-C391C	390	5	66	90	40	1.625 (41.28)	.500 (12.70)
CE1-C471C	470	5	73	75	40	1.125 (28.58)	.625 (15.88)
CE1-C681C	680	5	87	52	40	2.125 (53.98)	.500 (12.70)
CE1-C821C	820	5	96	43	40	1.125 (28.58)	.750 (19.05)
CE1-C122C	1200	5	116	29	40	1.125 (28.58)	.875 (22.23)
CE1-C152C	1500	5	130	23	40	1.625 (41.28)	.750 (19.05)
CE1-C182C	1800	5	142	19	40	2.625 (66.68)	.625 (15.88)
CE1-C332C	3300	5	193	8.5	40	1.625 (41.28)	1.000 (25.40)
CE1-C472C	4700	5	230	7.5	40	2.625 (66.68)	.875 (22.23)
CE1-C682C	6800	5	276	5	40	2.625 (66.68)	1.000 (25.40)
CE1-C103C	10000	5	336	3.5	40	3.625 (92.08)	1.000 (25.40)
CE1-C151D	150	10	58	187	35	1.125 (28.58)	.500 (12.70)
CE1-C301D	300	10	82	93	35	1.625 (41.28)	.500 (12.70)
CE1-C471D	470	10	104	60	35	2.125 (53.98)	.500 (12.70)
CE1-C681D	680	10	123	40	35	1.625 (41.28)	.625 (15.88)
CE1-C911D	910	10	143	31	35	2.125 (53.98)	.625 (15.88)
CE1-C102D	1000	10	150	28	35	2.125 (53.98)	.750 (19.05)
CE1-C152D	1500	10	183	19	35	2.125 (53.98)	.750 (19.05)
CE1-C222D	2200	10	222	13	35	1.625 (41.28)	1.000 (25.40)
CE1-C332D	3300	10	272	9	35	2.125 (53.98)	1.000 (25.40)
CE1-C472D	4700	10	325	6	35	2.625 (66.68)	1.000 (25.40)
CE1-C682D	6800	10	390	4	35	3.625 (92.08)	1.000 (25.40)
CE1-C101E	100	15	58	280	30	1.125 (28.58)	.500 (12.70)
CE1-C221E	220	15	86	127	30	1.625 (41.28)	.625 (15.88)
CE1-C361E	360	15	110	78	30	2.125 (53.98)	.500 (12.70)
CE1-C511E	510	15	130	55	30	1.625 (41.28)	.625 (15.88)
CE1-C821E	820	15	166	34	30	1.625 (41.28)	.750 (19.05)
CE1-C102E	1000	15	183	28	30	2.125 (53.98)	.750 (19.05)
CE1-C152E	1500	15	225	19	30	2.625 (66.68)	.750 (19.05)
CE1-C222E	2200	15	272	13	30	2.625 (66.68)	.875 (22.23)
CE1-C332E	3300	15	334	8.5	30	2.625 (66.68)	1.000 (25.40)
CE1-C472E	4700	15	398	6.0	30	3.625 (92.08)	1.000 (25.40)
CE1-C470F	47	25	51	314	25	1.125 (28.58)	.500 (12.70)
CE1-C101F	100	25	75	148	25	1.625 (41.28)	.500 (12.70)
CE1-C201F	200	25	106	74	25	2.125 (53.98)	.500 (12.70)
CE1-C271F	270	25	123	55	25	1.625 (41.28)	.625 (15.88)
CE1-C331F	330	25	136	45	25	1.125 (28.58)	.875 (22.23)
CE1-C391F	390	25	148	38	25	2.125 (53.98)	.625 (15.88)
CE1-C471F	470	25	162	31	25	1.625 (41.28)	.750 (19.05)
CE1-C681F	680	25	195	22	25	2.125 (53.98)	.750 (19.05)
CE1-C102F	1000	25	237	15	25	2.125 (53.98)	.875 (22.23)
CE1-C152F	1500	25	291	10	25	3.125 (79.38)	.875 (22.23)
CE1-C222F	2200	25	352	7	25	3.125 (79.38)	1.000 (25.40)
CE1-C272F	2700	25	390	5	25	3.625 (92.08)	1.000 (25.40)
CE1-C330G	33	50	61	450	20	1.125 (28.58)	.500 (12.70)
CE1-C680G	68	50	87	218	20	1.625 (41.28)	.500 (12.70)
CE1-C101G	100	50	106	148	20	2.125 (53.98)	.500 (12.70)
CE1-C151G	150	50	130	99	20	1.125 (28.58)	.875 (22.23)
CE1-C201G	200	50	150	74	20	2.125 (53.98)	.625 (15.88)
CE1-C241G	240	50	165	62	20	2.625 (66.68)	.625 (15.88)
CE1-C331G	330	50	192	45	20	1.625 (41.28)	.875 (22.23)

See footnotes at end of table.

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TABLE I. Styles CE12 and CE13 capacitors - continued.

PIN 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 +10, -5 Hz, -40°C +0°C, -3°C	Dissipation factor	Case dimensions 2/ 3/	
						A	B
						± .062 (1.57)	± .031 (0.79)
	µf	Volts	µa	Ohms	Percent	Inches (mm)	Inches (mm)
CE1-C471G	470	50	229	31	20	2.625 (66.68)	.750 (19.05)
CE1-C511G	510	50	240	29	20	2.125 (53.98)	.875 (22.23)
CE1-C681G	680	50	276	22	20	2.625 (66.68)	.875 (22.23)
CE1-C102G	1000	50	336	15	20	2.625 (66.68)	1.000 (25.40)
CE1-C152G	1500	50	411	10	20	3.625 (92.08)	1.000 (25.40)
CE1-C100H	10	100	48	880	15	1.125 (28.58)	.500 (12.70)
CE1-C180H	18	100	63	490	15	1.625 (41.28)	.500 (12.70)
CE1-C390H	39	100	93	226	15	1.625 (41.28)	.625 (15.88)
CE1-C620H	62	100	117	142	15	2.125 (53.98)	.625 (15.88)
CE1-C101H	100	100	150	88	15	2.125 (53.98)	.750 (19.05)
CE1-C111H	110	100	158	80	15	2.625 (66.68)	.750 (19.05)
CE1-C151H	150	100	183	59	15	2.125 (53.98)	.875 (22.23)
CE1-C221H	220	100	225	40	15	3.125 (79.38)	.875 (22.23)
CE1-C331H	330	100	272	27	15	3.125 (79.38)	1.000 (25.40)
CE1-C471H	470	100	325	19	15	3.625 (92.08)	1.000 (25.40)
CE1-C5R1J	5.1	150	42	1720	12	1.125 (28.58)	.500 (12.70)
CE1-C100J	10	150	58	880	12	1.625 (41.28)	.500 (12.70)
CE1-C200J	20	150	82	440	12	2.125 (53.98)	.500 (12.70)
CE1-C270J	27	150	96	326	12	1.625 (41.28)	.625 (15.88)
CE1-C330J	33	150	105	266	12	1.125 (28.58)	.875 (22.23)
CE1-C390J	39	150	114	226	12	2.125 (53.98)	.625 (15.88)
CE1-C470J	47	150	126	187	12	1.625 (41.28)	.750 (19.05)
CE1-C680J	68	150	151	129	12	1.625 (41.28)	.875 (22.23)
CE1-C910J	91	150	175	97	12	2.625 (66.68)	.750 (19.05)
CE1-C101J	100	150	183	88	12	2.125 (53.98)	.875 (22.23)
CE1-C151J	150	150	225	59	12	3.125 (79.38)	.875 (22.23)
CE1-C271J	270	150	302	33	12	3.625 (92.08)	1.000 (25.40)
CE1-C5R1K	5.1	200	51	1720	12	1.125 (28.58)	.500 (12.70)
CE1-C100K	10	200	67	880	12	1.625 (41.28)	.500 (12.70)
CE1-C150K	15	200	82	585	12	2.125 (53.98)	.500 (12.70)
CE1-C200K	20	200	95	440	12	1.625 (41.28)	.625 (15.88)
CE1-C300K	30	200	116	293	12	2.125 (53.98)	.625 (15.88)
CE1-C510K	51	200	150	172	12	2.125 (53.98)	.750 (19.05)
CE1-C750K	75	200	183	117	12	2.625 (66.68)	.750 (19.05)
CE1-C101K	100	200	211	88	12	2.625 (66.68)	.875 (22.23)
CE1-C151K	150	200	260	59	12	3.125 (79.38)	.875 (22.23)
CE1-C221K	220	200	314	40	12	3.625 (92.08)	1.000 (25.40)
CE1-C100M	10	250	75	880	12	2.125 (53.98)	.500 (12.70)
CE1-C150M	15	250	92	585	12	1.625 (41.28)	.625 (15.88)
CE1-C200M	20	250	106	440	12	2.125 (53.98)	.625 (15.88)
CE1-C220M	22	250	111	400	12	1.625 (41.28)	.750 (19.05)
CE1-C390M	39	250	148	226	12	2.125 (53.98)	.750 (19.05)
CE1-C470M	47	250	164	187	12	2.625 (66.68)	.750 (19.05)
CE1-C620M	62	250	186	142	12	2.125 (53.98)	.875 (22.23)
CE1-C101M	100	250	237	88	12	3.125 (79.38)	.875 (22.23)
CE1-C151M	150	250	290	59	12	3.625 (92.08)	1.000 (25.40)
CE1-C2R7N	2.7	300	43	3260	12	1.125 (28.58)	.500 (12.70)
CE1-C5R1N	5.1	300	58	1725	12	1.625 (41.28)	.500 (12.70)
CE1-C6R2N	6.2	300	64	1420	12	2.125 (53.98)	.500 (12.70)
CE1-C100N	10	300	82	880	12	1.625 (41.28)	.625 (15.88)
CE1-C200N	20	300	116	440	12	2.125 (53.98)	.625 (15.88)
CE1-C270N	27	300	135	326	12	2.625 (66.68)	.625 (15.88)

See footnotes at end of table.

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TABLE I. Styles CE12 and CE13 capacitors - continued.

PIN 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 +10, -5 Hz, -40°C +0°C, -3°C	Dissipation factor	Case dimensions 2/ 3/	
						A	B
						± .062 (1.57)	± .031 (0.79)
	µf	Volts	µa	Ohms	Percent	Inches (mm)	Inches (mm)
CE1-C360N	36	300	156	244	12	2.125 (53.98)	.750 (19.05)
CE1-C470N	47	300	178	187	12	2.625 (66.68)	.750 (19.05)
CE1-C680N	68	300	214	129	12	2.625 (66.68)	.875 (22.23)
CE1-C101N	100	300	260	88	12	3.125 (79.38)	1.000 (25.40)
CE1-C2R2P	2.2	350	41	4000	12	1.125 (28.58)	.500 (12.70)
CE1-C5R1P	5.1	350	63	1720	12	1.625 (41.28)	.500 (12.70)
CE1-C100P	10	350	88	880	12	1.625 (41.28)	.625 (15.88)
CE1-C150P	15	350	108	585	12	2.125 (53.98)	.625 (15.88)
CE1-C270P	27	350	145	326	12	2.125 (53.98)	.750 (19.05)
CE1-C390P	39	350	173	225	12	2.125 (53.98)	.875 (22.23)
CE1-C680P	68	350	232	129	12	2.625 (66.68)	1.000 (25.40)
CE1-C111P	110	350	294	80	12	3.625 (92.08)	1.000 (25.40)
CE1-C2R0Q	2.0	400	42	7400	17	1.125 (28.58)	.500 (12.70)
CE1-C4R3Q	4.3	400	62	3440	17	1.625 (41.28)	.500 (12.70)
CE1-C8R2Q	8.2	400	86	1800	17	1.625 (41.28)	.625 (15.88)
CE1-C9R1Q	9.1	400	90	1625	17	2.125 (53.98)	.625 (15.88)
CE1-C100Q	10	400	95	1480	17	1.125 (28.58)	.875 (22.23)
CE1-C150Q	15	400	116	986	17	2.625 (66.68)	.625 (15.88)
CE1-C200Q	20	400	133	740	17	2.125 (53.98)	.750 (19.05)
CE1-C240Q	24	400	147	616	17	2.625 (66.68)	.750 (19.05)
CE1-C390Q	39	400	187	380	17	2.625 (66.68)	.875 (22.23)
CE1-C470Q	47	400	203	314	17	3.125 (79.38)	.875 (22.23)
CE1-C680Q	68	400	248	218	17	3.125 (79.38)	1.000 (25.40)
CE1-C820Q	82	400	271	180	17	3.625 (92.08)	1.000 (25.40)
CE1-C1R0R	1.0	450	31	22000	17	1.125 (28.58)	.500 (12.70)
CE1-C3R0R	3.0	450	55	7330	17	1.625 (41.28)	.500 (12.70)
CE1-C5R1R	5.1	450	72	4310	17	1.625 (41.28)	.625 (15.88)
CE1-C6R2R	6.2	450	79	3520	17	2.125 (53.98)	.625 (15.88)
CE1-C100R	10	450	100	2200	17	2.625 (66.68)	.625 (15.88)
CE1-C150R	15	450	123	1465	17	2.125 (53.98)	.750 (19.05)
CE1-C220R	22	450	149	1000	17	2.125 (53.98)	.875 (22.23)
CE1-C330R	33	450	182	666	17	2.125 (53.98)	1.000 (25.40)
CE1-C470R	47	450	217	468	17	2.625 (66.68)	1.000 (25.40)
CE1-C680R	68	450	262	323	17	3.625 (92.08)	1.000 (25.40)

1/ Complete PIN shall include an additional digit to complete style.
2/ The dimensions shown are for style CE12; for style CE13, add .062 inch (1.57 mm) ± .062 inch (1.57 mm) to the nominal dimension.
3/ Metric equivalents are given for general information only.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case type: Tubular, uninsulated (CE12) and insulated (CE13).

Capacitance value: See table I.

Capacitance tolerance:

At 200 volts, dc rated, and above: -10, +50 percent.

At 150 volts, dc rated, and less: -10, +75 percent.

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Operating temperature range: -40°C to +85°C.

DC leakage (DCL): See table I.

Dissipation factor (DF): See table I.

Terminal strength: Test conditions A and D of MIL-STD-202-211 as specified in MIL-PRF-62.

Thermal shock and immersion:

Thermal shock: In accordance with MIL-PRF-62.

Immersion: In accordance with MIL-PRF-62.

Dielectric withstanding voltage (DWV) (insulating sleeves): In accordance with MIL-PRF-62.

DCL: Not more than 150 percent of initial measurement.

Capacitance change: 5 percent.

DF: See table I.

Corrosion: Not more than 10 percent.

Moisture resistance: In accordance with MIL-PRF-62.

Insulation resistance (insulating sleeve): Not less than 100 megohms.

DWV (insulating sleeves): In accordance with MIL-PRF-62.

DCL: Not more than 150 percent of initial requirement.

Capacitance change: 5 percent.

DF: See table I.

Corrosion: Not more than 10 percent.

Stability at low and high temperature: In accordance with MIL-PRF-62.

Step 1 (at +25°C):

DCL: See table I.

Capacitance: See table I.

DF: See table I.

Step 2 (at -40°C):

Impedance: See table I.

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Step 3 (at +25°C):

DCL: See table I.

Capacitance: ±5 percent of initial value.

DF: See table I.

Step 4 (at +85°C):

DCL: Not more than 10 times initial value, or meet initial +25°C requirement.

Capacitance: ±20 percent of initial value.

DF: See table I.

Step 5 (at +25°C):

DCL: See table I.

Capacitance: ±5 percent of initial value.

DF: See table I.

Surge voltage: In accordance with MIL-PRF-62.

DCL: See table I.

Capacitance change: 6 percent.

DF: See table I.

Vent: Not applicable.

Life: In accordance with MIL-PRF-62.

DCL: See table I.

Capacitance change:

100 volts, dc rated, and less: -20, +15 percent.

DC rated voltage above 100 volts: ±15 percent.

DF: Meet the initial requirement or not exceed 150 percent of the initial measurement.

Shelf life: In accordance with MIL-PRF-62.

Build factor: Not more than 4.0.

Shelf factor: Not more than 3.0.

DCL: Not more than 150 percent of initial value.

Capacitance change: 6 percent.

DF: Not more than 175 percent of initial requirement.

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Application data:

All capacitors shall withstand an rms ripple current at 120 Hz and +85°C as determined from the formula:

$$I_r \; k\sqrt{C}$$

Where:

I_r = rms ripple current in milliamps.
C = nominal capacitance in microfarads.
k = 20 for units rated at 5 to 250 volts dc inclusive.
= 30 for units rated at 300 to 450 volts dc inclusive.

Supersession data: See table II.

TABLE II. Supersession data.

Superseded insulated part number: M62/02-	Superseded uninsulated part number: M62/02-	Recommended replacement PIN (style CE12 or CE13)	Superseded insulated part number: M62/02-	Superseded uninsulated part number: M62/02-	Recommended replacement PIN (style CE12 or CE13)
001	101	CE1-C471C	039	139	CE1-C200K
002	102	CE1-C681C	040	140	CE1-C300K
003	103	CE1-C911D	041	141	CE1-C300K
004	104	CE1-C152C	042	142	CE1-C510K
005	105	CE1-C182C	043	143	CE1-C750K
006	106	CE1-C301D	044	144	CE1-C5R1N
007	107	CE1-C471D	045	145	CE1-C150M
008	108	CE1-C681D	046	146	CE1-C200M
009	109	CE1-C681D	047	147	CE1-C270N
010	110	CE1-C102D	048	148	CE1-C390M
011	111	CE1-C221E	049	149	CE1-C390M
012	112	CE1-C361E	050	150	CE1-C5R1N
013	113	CE1-C511E	051	151	CE1-C6R2N
014	114	CE1-C681F	052	152	CE1-C100N
015	115	CE1-C821E	053	153	CE1-C150P
016	116	CE1-C101F	054	154	CE1-C200N
017	117	CE1-C201F	055	155	CE1-C270P
018	118	CE1-C271F	056	156	CE1-C360N
019	119	CE1-C391F	057	157	CE1-C5R1P
020	120	CE1-C471F	058	158	CE1-C100P
021	121	CE1-C680G	059	159	CE1-C150P
022	122	CE1-C101G	060	160	CE1-C150P
023	123	CE1-C201G	061	161	CE1-C200Q
024	124	CE1-C201G	062	162	CE1-C270P
025	125	CE1-C241G	063	163	CE1-C2R0Q
026	126	CE1-C390H	064	164	CE1-C4R3Q
027	127	CE1-C620H	065	165	CE1-C8R2Q
028	128	CE1-C620H	066	166	CE1-C9R1Q
029	129	CE1-C101H	067	167	CE1-C150Q
030	130	CE1-C111H	068	168	CE1-C200Q
031	131	CE1-C100J	069	169	CE1-C240Q
032	132	CE1-C270J	070	170	CE1-C1R0R
033	133	CE1-C270J	071	171	CE1-C3R0R
034	134	CE1-C390J	072	172	CE1-C5R1R
035	135	CE1-C470J	073	173	CE1-C6R2R
036	136	CE1-C910J	074	174	CE1-C100R
037	137	CE1-C100K	075	175	CE1-C150R
038	138	CE1-C100K	076	176	CE1-C150R

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Referenced documents. In addition to MIL-PRF-62, this document references the following:

MIL-STD-202-211

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2020-018)

Review activities:
Army - AR, AV, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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